

# Primordial helium bounds on a leptonic electron-mass transition inside the nucleosynthesis window

Justin Pulford<sup>1,\*</sup>

<sup>1</sup>*Unaffiliated*

(Dated: August 2, 2026)

We treat the amplitude  $\varepsilon$  of a universal fractional shift in the electron mass that turns on inside the big-bang nucleosynthesis (BBN) temperature window as a free parameter, and ask what the measured primordial helium mass fraction  $Y_p$  implies for it. Bounds on constant or power-law variations of fundamental constants at BBN already exist in the literature; the narrow point here is only the temperature-dependent *turn-on* topology. The shift is switched on below a critical temperature  $T_c$  by a linear ramp,  $\varepsilon_{\text{eff}}(T) = \varepsilon \max(0, 1 - T/T_c)$ , which deforms the weak rates in the nuclear network without changing the relativistic degrees of freedom. At the single  $T_c$  where the helium response was measured in a production BBN code ( $T_c = 179 \text{ keV}$ ), that response is linear,  $dY_p/d\varepsilon = 0.00163$  per percent of  $\varepsilon$ , and reproduces a direct windowed run to a few percent of the window itself. Against the Aver *et al.* determination  $Y_p = 0.2453 \pm 0.0034$ , that elasticity implies

$$\varepsilon < 3.2\% \quad (2\sigma)$$

at that measured  $T_c$  only. The EMPRESS determination cannot be used for the same purpose: standard BBN with  $\varepsilon = 0$  already sits  $2.9\sigma$  above it, so a measurement in tension with the null hypothesis does not constrain a deformation of that null. Deuterium is not used for a derivative bound, because its  $\varepsilon$ -response is a nonlinear bottleneck quantity; no chain-dependent absolute D/H is quoted. The result is a single-observable, chain-free upper edge on  $\varepsilon$  for one ramp point, with zero fitted parameters — not a first bound on varying  $m_e$ , and not a scan over the full free- $T_c$  window.

## I. INTRODUCTION

A fractional change in the electron mass at early times is a standard deformation of big-bang nucleosynthesis and of recombination. In the microwave background it is partly degenerate with the expansion history and has been studied as a possible contribution to the Hubble tension [1–3]. At nucleosynthesis the same deformation acts through the weak rates and the binding-sensitive nuclear network, and the response matrices for varying constants have long been tabulated [4]. That literature already supplies order-of-magnitude bounds on constant or power-law shifts in  $m_e$  and related couplings; nothing below re-derives those bounds or claims priority over them.

Most existing BBN analyses assume a constant shift from some high temperature through the BBN window, or a power-law variation in redshift. The case of interest here is narrower: an electron-mass shift that is *absent* above a critical temperature  $T_c$  and turns on below it. That topology matters because neutron–proton freeze-out and the deuterium bottleneck sit at different temperatures ( $\sim 800 \text{ keV}$  and  $\sim 70 \text{ keV}$ , respectively). A transition that switches on between them deforms the two epochs unequally. The novelty claimed is only this topology applied to a production network at one measured  $T_c$ , not a first limit on varying  $m_e$ .

We do not propose a mechanism for such a transition, and we do not fix  $T_c$  or  $\varepsilon$  to any particular model value.

The question is purely kinematic: if an electron-mass shift of amplitude  $\varepsilon$  turns on inside the BBN window, what does the measured helium mass fraction allow? The answer is a single number against one observational determination (Aver *et al.*), at the single  $T_c$  where the helium elasticity was measured, together with an explicit statement that the other common determination (EMPRESS) cannot be used for this purpose.

Section II fixes the ramp and the free parameters. Section III reports the measured helium response. Section IV derives the Aver bound and states how a conservative  $T_c$  scan would extend it. Section V explains why EMPRESS yields no bound. Section VI states what is deliberately left out. Section VII summarises.

## II. SETUP

Let  $m_e^{(0)}$  be the present electron mass. We write a temperature-dependent fractional shift

$$m_e(T) = m_e^{(0)} [1 + \varepsilon_{\text{eff}}(T)], \quad (1)$$

with a linear turn-on below a critical temperature  $T_c$ ,

$$\varepsilon_{\text{eff}}(T) = \varepsilon \max(0, 1 - T/T_c). \quad (2)$$

Above  $T_c$  the shift is off; at  $T \rightarrow 0$  it saturates at the constant amplitude  $\varepsilon$ . Both  $\varepsilon$  and  $T_c$  are free. The only structural commitment is the ramp shape of Eq. (2), which is the minimal continuous switch-on that vanishes above  $T_c$  and is linear in the order-parameter sense below it. Nothing below depends on why the switch-on has that form.

---

\* pulfordj420@gmail.com

The relevant temperature window for a transition that BBN can actually witness is bounded below by the deuterium bottleneck ( $\sim 70$  keV) and above by the weak-rate window approaching neutron–proton freeze-out (of order a few hundred keV). We adopt the interval

$$T_c \in [70, 500] \text{ keV} \quad (3)$$

as the range over which  $T_c$  is regarded as free for the present purpose. Outside that interval the transition either misses the bottleneck entirely (too low) or reaches freeze-out in full (too high); both extremes change the interpretation of a helium-only bound and are not the case studied here.

The shift enters BBN through the electron mass in the weak rates. It does not add relativistic degrees of freedom: a network run with the ramp at fixed  $\varepsilon$  returns  $N_{\text{eff}}$  unchanged at the level of the code’s output [5]. The deformation is therefore a pure rate effect, not an expansion-rate lever.

### III. MEASURED HELIUM RESPONSE

The bound is built from a production run of the PRyMordial nuclear network [5], with the ramp of Eq. (2) applied pointwise to the weak rates: at each temperature the standard and fully shifted rates are blended by the weight  $w(T) = \max(0, 1 - T/T_c)$ . The run that defines the response was performed at

$$T_c = 179 \text{ keV}, \quad \varepsilon = 1.2543\%, \quad (4)$$

on the code’s default baryon density. The helium mass fraction moves from the  $\varepsilon = 0$  baseline to the windowed value as

$$Y_p^{(0)} = 0.246891 \longrightarrow Y_p^{(\varepsilon)} = 0.248995, \quad (5)$$

a fractional change of  $+0.852\%$ .

A four-point scan in  $\varepsilon$  at the same  $T_c$  establishes that the helium response is linear to a few percent over the range of interest, with elasticity

$$\frac{dY_p}{d\varepsilon} = 0.00163 \quad \text{per percent of } \varepsilon. \quad (6)$$

Multiplying Eq. (6) by  $\varepsilon = 1.2543\%$  recovers the window of Eq. (5) to 2.8% of the window itself. The linearity is what licenses a derivative bound; without it the translation from one windowed run into a limit on  $\varepsilon$  would not be legitimate.

Helium is almost insensitive to the baryon density,  $Y_p \propto \omega_b^{0.04}$ . Shifting  $\omega_b$  at the percent level therefore moves the baseline by far less than the observational error used below, so the bound is baseline-robust against the usual CMB–BBN  $\omega_b$  re-inference.

### IV. THE AVER BOUND

Aver *et al.* report [6]

$$Y_p^{\text{Aver}} = 0.2453 \pm 0.0034. \quad (7)$$

Writing  $Y_p(\varepsilon) = Y_p^{(0)} + 0.00163 \varepsilon\%$  with  $\varepsilon\%$  the amplitude in percent, and requiring  $Y_p(\varepsilon)$  to lie within  $n\sigma$  of Eq. (7), one finds

$$\varepsilon < 1.11\% \quad (1\sigma), \quad (8)$$

$$\varepsilon < 3.20\% \quad (2\sigma). \quad (9)$$

The  $2\sigma$  statement is the result of this paper:

$$\boxed{\varepsilon < 3.2\% \quad (2\sigma)} \quad (10)$$

against Aver, at the single  $T_c$  of Eq. (4), with no fitted parameters.

For orientation: the  $\varepsilon = 0$  baseline already sits  $+0.47\sigma$  above Aver, and the windowed point of Eq. (4) sits at  $+1.09\sigma$ . The transition is constrained but not excluded at the amplitude of Eq. (4). The  $1\sigma$  column of Eq. (8) is reported for completeness; the quotable bound is Eq. (10).

*a. Conservative  $T_c$  scan (method only; not produced here).* The free-parameter statement of Sec. II treats  $T_c$  as free on Eq. (3), but the verified number of Eq. (10) uses the helium elasticity measured at one point only. A conservative extension would re-measure  $dY_p/d\varepsilon$  on a grid of  $T_c$  values spanning Eq. (3), at fixed standard  $N_{\text{eff}}$  and fixed code baryon density, invert each elasticity against Aver as above, and report the upper envelope  $\varepsilon_{\text{max}}(T_c)$  — quoting only the most permissive (largest) ceiling if a single number is required, never a tighter bound at an unmeasured  $T_c$ . That grid is not reported here. The number in Eq. (10) is therefore the verified bound at the measured  $T_c$ ; it is not a claim that the same numerical ceiling holds at every point of Eq. (3), and it is not a claim that 179 keV maximises or minimises the helium response.

### V. WHY EMPRESS CANNOT BE USED

The EMPRESS collaboration reports a lower helium abundance [7],

$$Y_p^{\text{EMPRESS}} = 0.2370 \pm 0.0034. \quad (11)$$

Against that determination the same baseline and windowed values read

$$\varepsilon = 0 : +2.91\sigma, \quad \varepsilon = 1.2543\% : +3.53\sigma. \quad (12)$$

Standard BBN with no transition is already  $2.9\sigma$  high. The window contributes only  $0.62\sigma$  of the  $3.53\sigma$ . A measurement that is in tension with the null hypothesis cannot constrain a deformation of that null: removing

the transition entirely leaves  $2.9\sigma$  standing. Quoting an EMPRESS-based upper limit on  $\varepsilon$  would attribute to the transition a discrepancy that exists without it. There is therefore *no* EMPRESS-based  $\varepsilon$  bound in this note — not an unquoted number, but a non-result: EMPRESS does not bound the ramp.

The observational helium disagreement itself — Aver and EMPRESS differ by about  $1.7\sigma$  on the same quantity — is a property of the data, not of the ramp (the well-known primordial-helium tension). Until it is resolved, a helium-only bound on  $\varepsilon$  must be reported against Aver alone and must state why EMPRESS is not used. Aver is not selected because it is kinder to the ramp; it is selected because it is the determination against which the  $\varepsilon = 0$  null is still consistent at  $\lesssim 0.5\sigma$ , so an upper edge on a positive helium response is well posed.

## VI. WHAT IS NOT USED

*Deuterium.* The deuterium abundance responds to  $\varepsilon$ , but as a bottleneck residue rather than an equilibrium abundance. A uniform- $\varepsilon$  derivative under-predicts the windowed effect, and a four-point scan does not yield a linear elasticity that can be inverted into a clean upper limit of the kind in Eq. (10). No derivative-based deuterium bound is claimed. Absolute D/H comparisons that depend on a re-inferred baryon density from a cosmological fit (chain-dependent model predictions) are outside the scope of this note by design: they are not needed for Eq. (10), and they are not used.

*Extra radiation.* An additional relativistic density  $\Delta N_{\text{eff}}$  moves  $Y_p$  and D/H in the same direction. It is an expansion-rate lever, not a measurement of  $\varepsilon$ , and it is not floated here. The bound of Eq. (10) is the pure rate response at fixed standard  $N_{\text{eff}}$ .

*Lithium.* The same windowed run moves  ${}^7\text{Li}/\text{H}$  by  $+0.26\%$ , a percent-level shift that neither causes nor cures the cosmological lithium problem. Lithium is recorded as unaffected at the level relevant to Eq. (10) and is not used in the bound.

*Model-fixed  $T_c$  or  $\varepsilon$ .* No central value of  $T_c$  or  $\varepsilon$  is adopted as an input. The run of Eq. (4) fixes the elasticity; the bound is the upper edge that Aver allows.

## VII. CONCLUSION

A universal electron-mass shift that turns on inside the BBN window, with free amplitude  $\varepsilon$  and free critical temperature  $T_c$  in the nucleosynthesis-relevant range, is bounded by primordial helium at the single  $T_c$  where the helium elasticity was measured. Against Aver *et al.*,

$$\varepsilon < 3.2\% \quad (2\sigma),$$

with zero fitted parameters. That is a chain-free, single-observable upper edge for one ramp point — not a first bound on varying  $m_e$ , and not a dense  $\varepsilon_{\text{max}}(T_c)$  map. The EMPRESS determination cannot be used for this purpose, because standard BBN is already  $2.9\sigma$  high against it before any transition is applied. Deuterium is excluded from the bound because its  $\varepsilon$ -response is non-linear, and no chain-dependent absolute D/H is quoted.

- 
- [1] L. Hart and J. Chluba, New constraints on time-varying fundamental constants from Planck, *Mon. Not. R. Astron. Soc.* **474**, 1850 (2018); arXiv:1705.03925.
  - [2] L. Hart and J. Chluba, Updated fundamental constant constraints from Planck 2018 data and possible relations to the Hubble tension, *Mon. Not. R. Astron. Soc.* **493**, 3255 (2020); arXiv:1912.03986.
  - [3] T. Sekiguchi and T. Takahashi, Early recombination as a solution to the  $H_0$  tension, *Phys. Rev. D* **103**, 083507 (2021); arXiv:2007.03381.
  - [4] T. Dent, S. Stern, and C. Wetterich, Primordial nucleosynthesis as a probe of fundamental physics parameters, *Phys. Rev. D* **76**, 063513 (2007); arXiv:0705.0696.
  - [5] A.-K. Burns, T. M. P. Tait, and M. Valli, PRyMordial: the first three minutes, within and beyond the Standard Model, *Eur. Phys. J. C* **84**, 86 (2024); arXiv:2307.07061.
  - [6] E. Aver, D. A. Berg, K. A. Olive, R. W. Pogge, J. J. Salzer, and E. D. Skillman, Improving helium abundance determinations with Leo P as a case study, *JCAP* **03**, 027 (2021); arXiv:2010.04180.
  - [7] A. Matsumoto *et al.* (EMPRESS Collaboration), EMPRESS. VIII. A new determination of primordial helium abundance with extremely metal-poor galaxies: a suggestion of the lepton asymmetry and implications for the Hubble tension, *Astrophys. J.* **941**, 167 (2022); arXiv:2203.09617.